OXFORD
INTERNATIONAL
AQA EXAMINATIONS

Please write clearly in block capitals.

Centre number | | | | | |    Candidate number | | | | |

Surname ____________________

Forename(s) ____________________

Candidate signature ____________________

# INTERNATIONAL A-LEVEL
# BIOLOGY (9610)

## Unit 3 Populations and genes

Tuesday 5 June 2018 07:00 GMT Time allowed: 1 hour 30 minutes

**Materials**
For this paper you must have:
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

**Instructions**
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 75.

| For Examiner's Use | |
|---|---|
| Question | Mark |
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |
| **TOTAL** | |

Answer **all** questions in the spaces provided.

**0 1** Lemmings are small herbivorous mammals. A population of lemmings lives in the Arctic ecosystem.

**0 1 . 1** Give the meaning of the terms:

**[2 marks]**

Population ____________________

____________________

____________________

Ecosystem ____________________

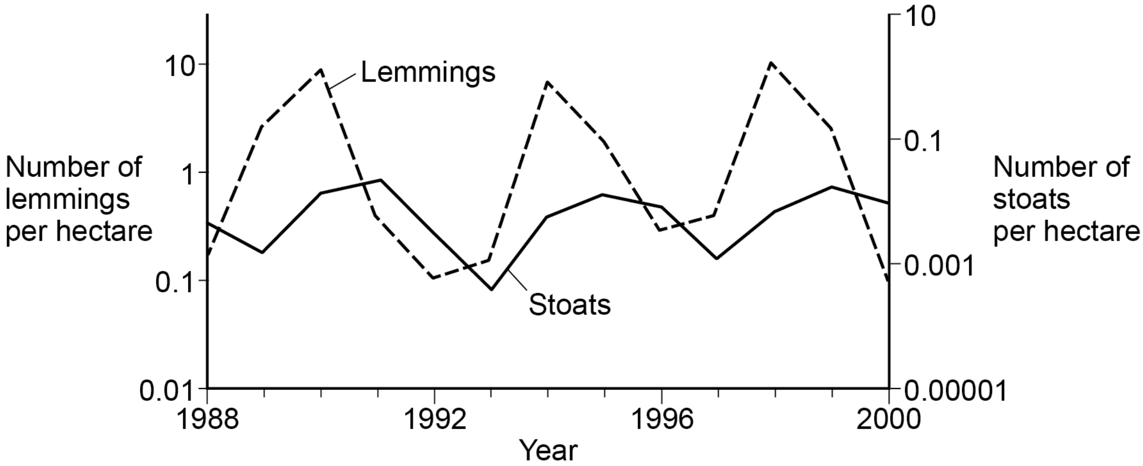
____________________

____________________

Stoats are carnivorous mammals that feed almost exclusively on lemmings.

**Figure 1** shows the changes in the numbers of lemmings and stoats in one region from 1988 to 2000.

**Figure 1**



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**0 1 . 2** Give the reason why logarithmic scales have been used on the y-axes in **Figure 1**.

**[1 mark]**

**0 1 . 3** Describe the relationship between the number of lemmings and the number of stoats.

Use information from **Figure 1**.

**[2 marks]**

**0 1 . 4** Lemmings feed on leaves, shoots, roots and bulbs. Predators of lemmings include stoats, owls and foxes.

A student suggests that an increase in the stoat population causes a decrease in the lemming population.

Evaluate this statement.

Use the information provided above and in **Figure 1**.

**[4 marks]**

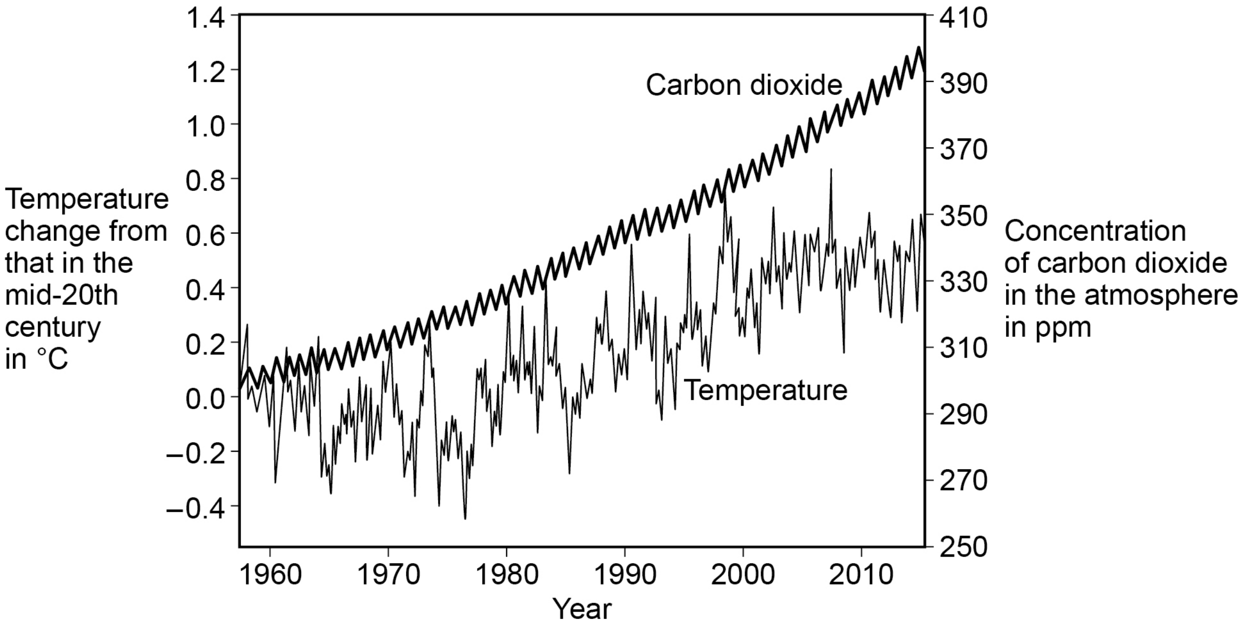
9

**0 2** The Mauna Loa Observatory in Hawaii has been collecting and monitoring data about climate change since 1956.

The equipment and techniques used to collect the data have remained unchanged during the monitoring period.

**Figure 2** shows changes in atmospheric carbon dioxide concentration and in air temperature since 1956.

**Figure 2**




**0 2 . 1** Give **one** reason why the equipment and techniques used to collect the data shown in **Figure 2** have been kept the same.

**[1 mark]**

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**0 2 . 2** **Figure 2** shows a regular fluctuation in the carbon dioxide concentration every year.

Suggest **one** reason for the fluctuation.

**[2 marks]**

**0 2 . 3** Do the data in **Figure 2** support the idea that an increase in atmospheric carbon dioxide concentration has caused an increase in global temperature?

Explain your answer.

**[3 marks]**

**6**

**Turn over for the next question**

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**0 3** Flower colour in the *Salvia* plant is controlled by two genes.

One gene determines the production of a pigment. The other gene determines the colour of the pigment. The two genes are not linked.

Each gene has two alleles.

The dominant allele **A** causes a pigment to be produced. Plants with the genotype **aa** will produce no pigment and the flowers will be white.

The dominant allele **B** produces a plant with purple flowers. Plants with the genotype **bb** will have pink flowers.

**0 3 . 1** Name the interaction between the two genes for flower colour.
Give a reason for your choice.

**[2 marks]**

Interaction ____________________

Reason ____________________

____________________

____________________

**0 3 . 2** Give **all** possible genotypes of a pink-flowered plant.

**[1 mark]**

____________________

____________________

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**0 3 . 3** A plant breeder crossed a purple-flowered plant heterozygous for both genes with a white-flowered plant homozygous recessive for both genes.

The plant breeder expected pink, purple and white flowers in the ratio 1 : 1 : 2

Draw a genetic diagram to show how this ratio of phenotypes can be produced.

**[3 marks]**

| | | |
|---|---|---|
| Parent phenotypes | Purple flowers | White flowers |
| Parent genotypes | | |
| Gametes | | |
| Offspring genotypes | | |
| Offspring phenotypes | | |

**Question 3 continues on the next page**

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**0 3 . 4** The plant breeder counted the number of plants with each flower colour and compared them with the expected ratio.

The plant breeder completed a statistical test to see if the difference in the observed and expected ratio was due to chance.

**Table 1** shows values for $\chi^2$ (chi squared) at different probability levels and for different degrees of freedom.

**Table 1**

| Degrees of freedom | Probability, p | | | | |
|---|---|---|---|---|---|
| | **0.2** | **0.1** | **0.05** | **0.02** | **0.01** |
| 1 | 1.64 | 2.71 | 3.84 | 5.41 | 6.64 |
| 2 | 3.22 | 4.61 | 5.99 | 7.82 | 9.21 |
| 3 | 4.64 | 6.25 | 7.82 | 9.84 | 11.35 |
| 4 | 5.99 | 7.78 | 9.49 | 11.67 | 13.28 |
| 5 | 7.29 | 9.24 | 11.07 | 13.39 | 15.09 |

The value of $\chi^2$ was calculated and found to be 4.73.

Explain what the results of the $\chi^2$ test show about the difference between the observed and expected ratio.

Use **Table 1** and the calculated value of $\chi^2$.

**[2 marks]**

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**0 3 . 5** A population of *Salvia* plants contains 49% white flowers.

Calculate the percentage of this population that you would expect to be heterozygous for the production of pigment gene.

Use the Hardy-Weinberg equation.

**[2 marks]**

Answer = ____________________

**10**

**Turn over for the next question**

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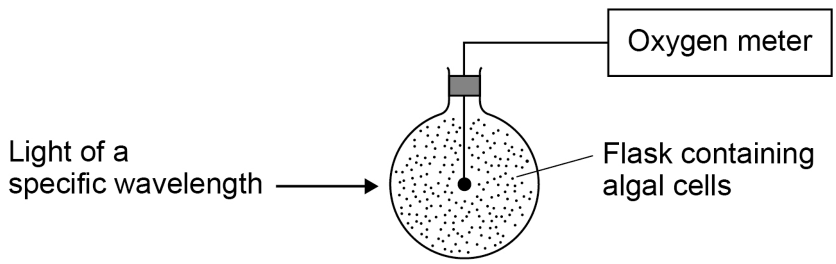
**0 4** A student investigates the effect of different wavelengths of light on the rate of photosynthesis.

The student:

- uses photosynthesising algal cells
- sets up the apparatus shown in **Figure 3**
- shines different wavelengths of light onto the flask
- takes measurements to determine the rate of photosynthesis.

**Figure 3**




**0 4 . 1** What **two** measurements should the student take to determine the rate of photosynthesis?

**[1 mark]**

______________________________________________________________________

______________________________________________________________________

**0 4 . 2** The student controls the temperature, pH and carbon dioxide concentration.

Give **two** other factors that the student should keep constant during this investigation.

**[2 marks]**

1 ____________________________________________________________________

______________________________________________________________________

2 ____________________________________________________________________

______________________________________________________________________

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**0 4 . 3** The student adds a reagent to the suspension of photosynthesising algal cells.

The reagent is blue when oxidised and is colourless when reduced.

The blue colour disappears when the suspension of algal cells is exposed to light.

Explain why.

Use your knowledge of the light-dependent reaction of photosynthesis.

**[2 marks]**

**0 4 . 4** Some purple bacteria can photosynthesise, but use hydrogen sulfide instead of water. The hydrogen sulfide has a similar role to water in photosynthesis.

Suggest how the bacteria use hydrogen sulfide.

**[3 marks]**

**0 4 . 5** Suggest how the products of the light-dependent reaction in purple bacteria would be different from those in photosynthetic algae.

**[1 mark]**

**9**

**0 5** Modern farming practices increase productivity in animals reared for meat and eggs.

A scientist investigates the effect of temperature on the rate of chicken growth and the efficiency of food conversion to biomass. The scientist uses the same breed of chicken for each temperature.

**Table 2** shows the results of the investigation.

**Table 2**

| Temperature / °C | Mean growth rate / g per day | Efficiency of conversion of food to biomass / % |
|---|---|---|
| 5 | 26 | 15 |
| 10 | 39 | 41 |
| 20 | 43 | 47 |
| 30 | 23 | 34 |
| 35 | 16 | 34 |

**0 5 . 1** How could the scientist calculate the efficiency of conversion of food to biomass?

**[1 mark]**

______________________________________________________________________

______________________________________________________________________

**0 5 . 2** A student interprets the results in **Table 2** and concludes that 20 °C is the optimum temperature for chicken growth.

Evaluate this conclusion.

**[2 marks]**

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**0 5 . 3** Explain why there is a higher efficiency of conversion of food to biomass at 20 °C than at 5 °C.

**[2 marks]**

**0 5 . 4** Suggest why the scientist did not use a temperature above 35 °C.

Use data from **Table 2**.

**[2 marks]**

**Question 5 continues on the next page**

Chickens are farmed for meat and egg production.

In some countries, chickens are kept in cages to increase the efficiency of conversion of food to biomass.

**Figure 4** shows a typical cage system for intensively farmed chickens.

**Figure 4**

| **Dimensions of a chicken** | **Guidelines for typical cage system** |
|---|---|
| Length of chicken = 30–45 cm<br>Wing span = 45–60 cm<br>Height = 35–45 cm | • cages should provide at least 750 $cm^2$ floor space per chicken<br>• cage height should be at least 45 cm |

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**0 5 . 5** Evaluate the use of cages for intensive chicken farming.

Use information from **Figure 4**.

**[3 marks]**

**10**

**Turn over for the next question**

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**0 6** Glucose is the main respiratory substrate used by cells.
The overall equation for the aerobic respiration of glucose is:

$$C_6H_{12}O_6 + 6\,O_2 \rightarrow 6\,CO_2 + 6\,H_2O$$

Oleic acid is a fatty acid. Fatty acids can be used as respiratory substrates.
The overall equation for the aerobic respiration of oleic acid is:

$$2\,C_{18}H_{34}O_2 + 51\,O_2 \rightarrow 36\,CO_2 + 34\,H_2O$$

**0 6 . 1** Calculate the ratio of the respiratory quotient (RQ) for the aerobic respiration of oleic acid to the RQ for the aerobic respiration of glucose.

**[2 marks]**

Ratio = ____________________

**0 6 . 2** Suggest one reason why an experimental RQ value does not always indicate the type of substrate used in respiration.

**[1 mark]**

______________________________________________________________________

______________________________________________________________________

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**Table 3** shows the relative energy values of carbohydrate and lipid.

**Table 3**

| **Respiratory substrate** | **Mean energy value / kJ $g^{-1}$** |
|---|---|
| Carbohydrate | 15.8 |
| Lipid | 39.4 |

**0 6 . 3** Glucose is a carbohydrate with the chemical formula $C_6H_{12}O_6$

Oleic acid is a breakdown product of lipids. The chemical formula for oleic acid is $C_{18}H_{34}O_2$

Explain why cells gain more energy from lipid than from carbohydrate.

Use your knowledge of aerobic respiration.

**[3 marks]**

**Question 6 continues on the next page**

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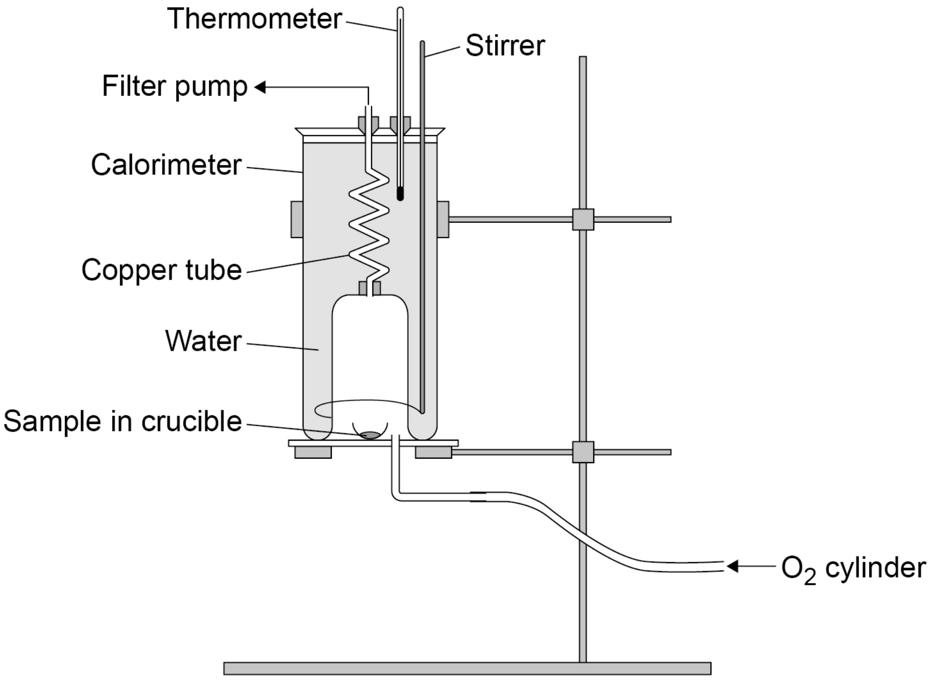
The energy value of a food can be found by burning the food and measuring the heat energy released.

A student uses a piece of equipment called a calorimeter to determine the mean energy value of glucose.

**Figure 5** shows a calorimeter.

**Figure 5**




**0 6 . 4** The student:

- adds a known mass of water to the calorimeter
- puts the glucose sample in the crucible
- sets the glucose on fire.

Suggest **two** measurements the student would take to determine the energy content of glucose in kJ $g^{-1}$

**[2 marks]**

1 ______________________________________________

______________________________________________

2 ______________________________________________

______________________________________________

8

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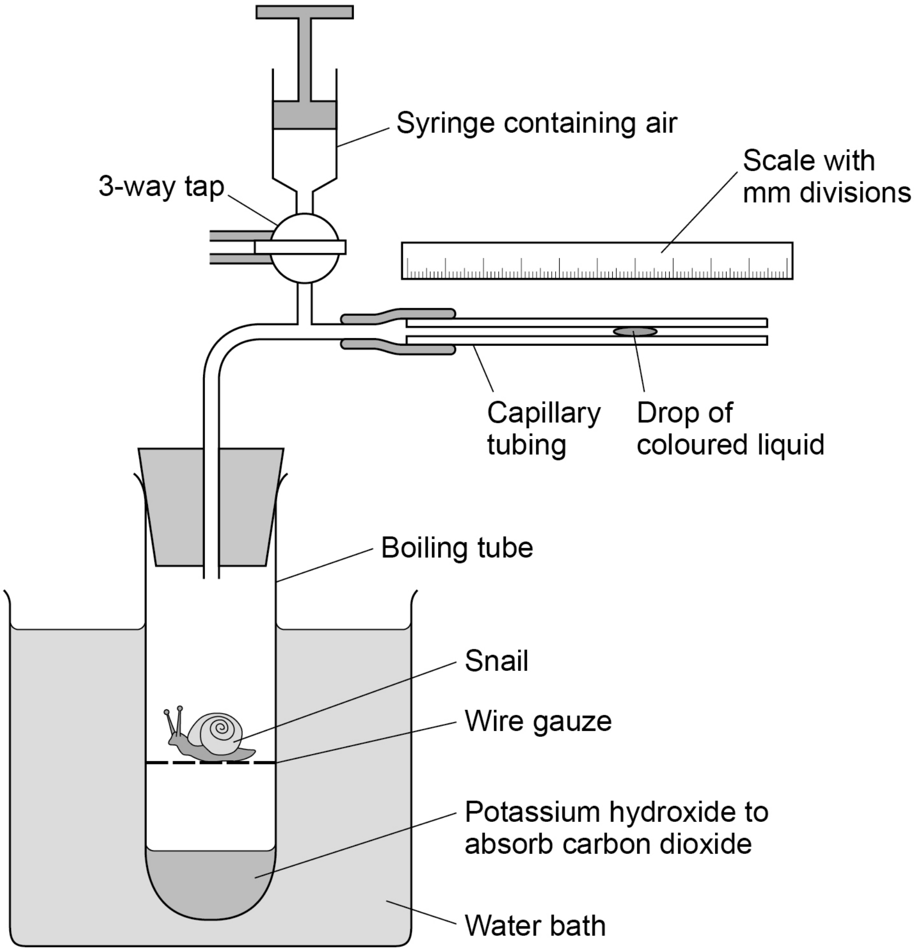
**0 7** A student investigates the effect of temperature on the rate of respiration of a snail.
The student uses the apparatus shown in **Figure 6**.

**Figure 6**




**0 7 . 1** Explain what would happen if the student replaced the potassium hydroxide with water.

**[2 marks]**

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**0 7 . 2** The internal diameter of the capillary tubing is 1 mm
The drop of coloured liquid moves 30 mm in 10 minutes
The snail has a mass of 7 g

Calculate the rate of oxygen uptake in $mm^3\ g^{-1}\ hour^{-1}$

Volume of a cylinder = $\pi r^2 h$

Use $\pi$ = 3.14. Give your answer to 3 significant figures.

**[3 marks]**

Rate of oxygen uptake = ____________________ $mm^3\ g^{-1}\ hour^{-1}$

**Question 7 continues on the next page**

A student investigates the effect of temperature on the rate of oxygen uptake in seashore snails. Seashore snails are covered with sea water at high tide and exposed to the air at low tide.

The student:

- measures the oxygen uptake of snails kept at different temperatures in moist air
- repeats the experiment using the same snails in different temperatures of sea water
- calculates the means and standard deviations for each temperature.

The student's results are shown in **Table 4**

**Table 4**

| | **Mean oxygen uptake of snails / $mm^3\ g^{-1}\ h^{-1}$ ($\pm$ standard deviation)** | |
|---|---|---|
| **Temperature / °C** | **In moist air** | **In seawater** |
| 5 | $35 \pm 2$ | $28 \pm 8$ |
| 10 | $34 \pm 6$ | $32 \pm 3$ |
| 15 | $36 \pm 3$ | $35 \pm 3$ |
| 20 | $86 \pm 8$ | $52 \pm 10$ |
| 25 | $141 \pm 13$ | $96 \pm 15$ |
| 30 | $132 \pm 14$ | $108 \pm 9$ |
| 35 | $120 \pm 16$ | $79 \pm 21$ |

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**0 7 . 3** The student concludes that 25 °C is the optimum temperature for the rate of oxygen uptake.

Evaluate this conclusion.

**[3 marks]**

**8**

**Turn over for the next question**

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**0 8** Open pit mining is a technique used to extract copper ore near the surface of the earth.

After mining finishes, the pit is filled in and the soil is replaced.

**0 8 . 1** Over time, succession occurs and the ecosystem around an old copper mine changes.

Describe and explain these changes.

**[6 marks]**

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**0 8 . 2** Soil around old copper mines has a high concentration of copper ions.
Copper ions are toxic to many plant species.
Some species of grass have developed a high tolerance to copper ions and grow readily in the soil around copper mines.

Explain how natural selection could produce a population of copper-tolerant grass.
**[4 marks]**

**Question 8 continues on the next page**

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**0 8 . 3** Copper-tolerant grass produces its flowers earlier in the year than non-tolerant grass of the same species.

Explain how this might produce two different species of grass.

**[5 marks]**

15

**END OF QUESTIONS**

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